

Task-Oriented Deep Joint Source–Channel Coding

Semantic communication for bandwidth-constrained image classification

The project studies whether an edge device can send the information needed for a classification task without having to transmit and reconstruct every image detail.

Capstone First Review

Review window: 18–22 August 2026

Presentation slot: 15 minutes

Team and guide details to be added

The project in one minute

1

Observe

An edge camera sees an image.

2

Send

The link has limited bandwidth and added noise.

3

Decide

A receiver predicts the image class.

4

Compare

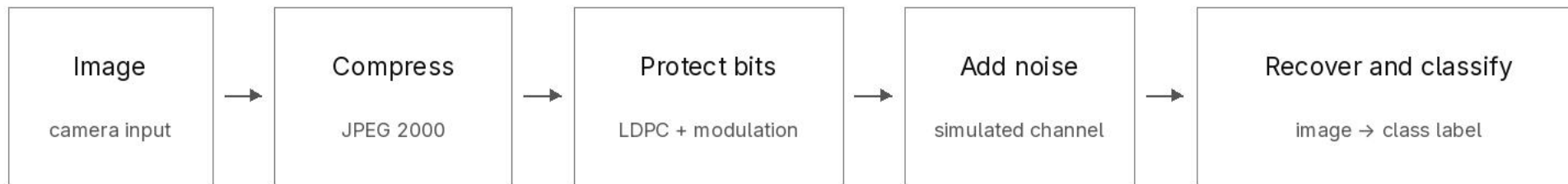
We measure task accuracy at the same channel budget.

Semantic communication

Send the information required to complete the task. Reconstructing every original pixel is optional when the task does not need it.

The final result is classification accuracy after transmission, not image quality by itself.

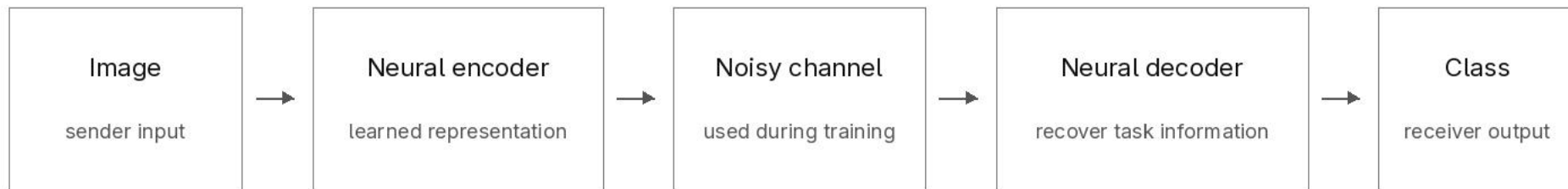
How a conventional image link solves the task



Compression	reduces the number of source bits.
Error correction	adds controlled redundancy so damaged bits can be recovered.
Modulation	maps protected bits to symbols sent through the channel.
Classifier	uses the recovered image to choose a class label.

Limitation: each stage is designed mainly to recover images and bits, while the final metric is whether the class decision is correct.

What the learned joint system changes



The central change

The sender and receiver are trained together through the channel. The transmitted representation can therefore preserve information useful for classification after noise.

Why not send the label?

The project fixes the deployment split: the sender has an encoder, while the receiver owns the decoder and classifier. Sending a label would move the full task model to the sender and test a different system.

The method is supervised end-to-end learning. It is not reinforcement learning and it does not use an LLM.

Why the comparison needs three systems

System	What it sends	What the comparison tells us
Classical image link	A compressed image protected by LDPC and modulation.	How a strong conventional separated system performs.
Digital feature control	Quantized learned features through the same digital physical layer.	Whether task-aware representation alone explains an improvement.
Learned joint link	A continuous representation learned together with the noisy channel.	Whether joint source–channel learning adds further value.

Research question At the same bandwidth and channel conditions, can the learned joint system preserve classification accuracy better than the tuned classical system, and is any difference specifically consistent with joint coding?

What prior research establishes—and leaves open

Short-packet
communication

Finite packets pay coding overhead and can fail abruptly.

Measure the practical baseline instead of assuming ideal capacity. [1–4]

Learned image
compression

Neural transforms can improve rate–distortion performance.

Good reconstruction quality is not the same as good task accuracy. [11–17]

DeepJSCC

Learned continuous mappings can degrade gradually under noise.

Most image studies emphasize reconstruction rather than classification. [5–10,24]

Task-oriented
communication

Learned features can preserve inference under link limits.

Task awareness and joint coding are often not isolated from each other. [18–23]

Gap addressed

A fair three-way comparison for image classification at equal channel use, with failures retained and one sealed test campaign.

How one image becomes one fair comparison

Held equal for all systems

Same image and dataset split

Why the control exists

Prevents easier samples from being assigned to one system.

Same complex-symbol budget

Makes bandwidth use directly comparable.

Same SNR definition and keyed noise

Pairs outcomes under the same channel condition.

Validation-only tuning

Keeps all choices away from the sealed test split.

Every failure stays in the denominator

Prevents failed transmissions from disappearing from accuracy.

Full byte and physical-layer identity

Prevents hidden overhead or missing BLER evidence from helping a system.

The test split remains sealed until gate G-12. No test result exists at First Review.

What the project will complete and test

Completion objectives

1. Build all three transmission systems.
2. Match their communication resources and channel conditions.
3. Freeze every choice before test access.
4. Run one paired test campaign and report the result either way.

Preregistered questions

- H1 At low SNR, does the learned system outperform the adaptive classical system?
- H2 Does a fixed classical configuration show a cliff while the learned system degrades more gradually?
- H3 Does the learned–classical gap contract as SNR improves?
- H4 Does the learned joint system also exceed the digital feature control?

Completion means executing the protocol correctly. A positive learned-system result is not required.

What has been demonstrated so far

Gate	Observed evidence	Plain-language meaning	Boundary
G-1	89.8% validation accuracy	The reference image-classification task is viable.	Validation only; test is sealed.
G-7	1.64 M parameters; measured GPU profile	The planned learned model fits the available GPU.	A feasibility profile, not training.
G-2	Golden vectors and BLER conformance pass	The digital physical-layer implementation behaves as expected.	Small conformance evidence, not the full table.
W4	Bounded conventional pipeline runs end to end	Compression, coding, channel, decoding and records are integrated.	Integration evidence, not a comparison result.
G8_C	C2 characterization paused at a durable checkpoint	Full-strength BLER measurement can resume safely.	Table, selection and training are unfinished.

No worker is running. No selection, training, validation decoding or test access has occurred.

What remains after First Review

Stage	State	Work	Output
G8_C → G-8	Paused → planned	Finish BLER characterization; freeze the table; select classical operating points.	Frozen baseline
W5–W8	Planned	Train DJSCC and calibrate the task/reconstruction objective.	Frozen learned model
W9	Planned	Build and verify the task-aware digital feature control.	Attribution control
W10–W12	Planned	Validation rehearsal; freeze all choices; run one sealed test campaign.	Hypothesis decisions
W13–W17	Planned	Prepare demo, report, poster and final review package.	Final submission

First Review: 18–22 Aug · Second Review: 29 Sep–3 Oct · Final Review: 17–21 Nov · Report due: 20 Nov

W16 is report contingency, not room for new experimental scope.

What this First Review should confirm

Research boundary	Compare the three systems on image classification over normalized AWGN at matched complex-symbol budgets, using one sealed test campaign.
Technical claim boundary	The conventional path uses OpenJPEG 2.5.4 and TS 38.212-derived LDPC/rate matching. The project does not claim a complete 5G NR radio link.
Deployment boundary	Simulation-first Tier 1 is the required path. Conducted SDR replay and edge-device deployment remain optional stretch work after G-5.
Decision requested	Confirm the task definition, the three-system comparison, the fairness controls and simulation-first Tier 1 as a sufficient capstone scope.

Outstanding human item: record the guide's dated acknowledgement of the no-required-hardware path.

Formal H1 rule and First Review coverage

H1 — low-SNR separation (confirmatory primary)

Comparator: learned versus classical_adaptive at the headline ratio. A grid point qualifies when the studentized paired mean $\sqrt{N} \cdot \Delta(s) / \sigma(s)$ exceeds 1.96. Let R_{obs} be the longest run of consecutive qualifying points at or below the training SNR. H1 is supported only if $R_{\text{obs}} \geq 3$ and the calibrated run p-value ≤ 0.05 .

The run rule decides H1. The mean paired accuracy difference over the full low-SNR region is the effect size of record. A curve crossing is reported if observed but is not required.

Outcome rule: support, no support, or an adverse result are all valid project findings. The protocol does not change after test access.

Six-criterion rubric map

Motivation	2–4
Objectives	1–2, 5, 8, 11
Hypothesis	8, 12
Problem Survey	6
Subject Knowledge	3–7, 9, 11–12
Time Plan	10

Backing evidence: 30-source literature review, corrected Gantt, standards register, deployment dossier and authenticated gate records.